Econometrics I
PPS-01
Multiple Choice Questions
(chapters 1-2-3) Prof. Dr. Hüseyin Taştan

1. **Which of the following model <u>is linear</u> in parameters?**

   A.
   $$y = \beta_0 + x^{\beta_1} + u$$

   B.
   $$\log(y) = \beta_0 + \beta_1 \log(x) + u$$

   C.
   $$y = \beta_0 + e^{\beta_1 x} + u$$

   D.
   $$y = \beta_0 + \frac{1}{\beta_1} x + u$$

   E.
   $$y = \beta_0 + \beta_1^2 x + u$$

2. **Which of the following model <u>is not linear</u> in parameters?**

   A.
   $$\log(y) = \beta_0 + \beta_1 \log(x) + u$$

   B.
   $$\log(y) = \beta_0 + \beta_1 x + u$$

   C.
   $$\log(y) = \beta_0 + \beta_1 x^2 + u$$

   D.
   $$y = \beta_0 + \beta_1^2 x + u$$

   E.
   $$y = \beta_0 + \beta_1 \log(x) + u$$

3. Consider the following data set:

colGPA	hsGPA
3	3
2	3
4	3
1	3
3	3

   Consider the regression of $colGPA = \beta_0 + \beta_1\ hsGPA + u$. Find the OLS estimate for β_1.

   A. $\hat{\beta}_1 = 0.5$
   B. $\hat{\beta}_1 = 1.5$
   C. $\hat{\beta}_1 = 1$
   D. $\hat{\beta}_1 = 0.2$
   E. $\hat{\beta}_1$ is undefined since there is no variation in hsGPA.

4. Consider the following data set:

colGPA	hsGPA
3	2
4	3
3	2
4	3
4	4

Consider the regression of $colGPA = \beta_0 + \beta_1\ hsGPA + u$. Find the OLS estimate for β_1.

A. $\hat{\beta}_1 = 1$

B. $\hat{\beta}_1 = 1.571$

C. $\hat{\beta}_1 = 0957$

D. $\hat{\beta}_1 = 0.571$

E. $\hat{\beta}_1 = 0.2154$

5. **Consider the following regression result in equation form:**

$$\widehat{\log(y)} = 1.5 + 0.80\log(x) + 0.50z$$

Which of the following statements is correct?

A. Given $\Delta z = 1$, then y is predicted to increase by 0.5%, ceteris paribus.

B. Given $\Delta z = 1$, then y is predicted to increase by 5%, ceteris paribus.

C. Given $\Delta z = 1$, then y is predicted to increase by 50%, ceteris paribus.

D. Given $\Delta z = 1$, then y is predicted to increase by 0.05%, ceteris paribus.

E. Given $\Delta z = 10$, then y is predicted to increase by 50%, ceteris paribus.

6. **Exclusion of a relevant variable from a multiple linear regression model leads to the problem of ..... .**

A. perfect collinearity

B. homoskedasticity

C. heteroskedasticity

D. omitted variable bias

E. no perfect collinearity

7. **Consider the following regression results:**

$$\widehat{\mathrm{y}} = \underset{(0.5)}{10.20} + \underset{(0.040)}{0.08}\ \mathrm{x}_1 - \underset{(0.5)}{0.50}\,\mathrm{x}_2 + \underset{(0.03)}{0.30}\,\mathrm{x}_3$$

$$n = 104 \quad R^2 = 0.50 \quad \hat{\sigma} = 0.5$$

(standard errors are in parentheses. $\hat{\sigma}$ is the standard error of the regression (SER).)

What is the total sum of squares (SST)?

A. 0.50

B. 0.25

C. 50

D. 25

E. SST cannot be found from the information given.

8. Consider the following data set:

hsGPA	colGPA
2	3
3	4
2	3
3	4
3	4

OLS solution is $\widehat{colGPA} = 1 + hsGPA$. Find R^2.

A. R^2 cannot be found from the given information.

B. $R^2 = 1$

C. $R^2 = 0$

D. $R^2 = 0.5$

E. $R^2 = -1$

9. Consider the following data set:

colGPA	hsGPA
3	3
2	3
4	3
3	3
3	3

We run the regression of $colGPA = \beta_0 + u$. Note that there is no explanatory variable. Find the OLS estimate for β_0.

A. $\hat{\beta}_0 = 1$

B. $\hat{\beta}_0 = 2$

C. $\hat{\beta}_0 = 3$

D. $\hat{\beta}_0 = 4$

E. $\hat{\beta}_0$ is undefined since there is no variation in hsGPA.

10. Consider the following data set:

x	y
-6	-3
-2	-2
0	0
2	2
6	3

We run the regression of $Y = \beta_1 X + u$. Note that this is regression through the origin. Find the OLS estimate for β_1.

A. $\hat{\beta}_1 = 0.40$

B. $\hat{\beta}_1 = 0.50$

C. $\hat{\beta}_1 = 0.60$

D. $\hat{\beta}_1 = 0.55$

E. $\hat{\beta}_1$ is undefined because there is no intercept.

11. Fill in the blanks. $\hat{\sigma} = \sqrt{SSR/dof}$ is also known as ..... . If $\hat{\sigma}$ increases, then the standard errors of OLS estimates .....

    A. the degrees of freedom / increase.

    B. the variance of the regression / increase.

    C. the standard error of the regression (SER) / become negative.

    D. the standard error of the regression (SER) / decrease.

    E. the standard error of the regression (SER) / increase.

12. Fill in the blank. **The value of coefficient of determination (R-squared) .....**

    A. always decreases or remains the same when new variables are added.

    B. always remains the same when new variables are added.

    C. sometimes decreases when new variables are added.

    D. always increases or remains the same when new variables are added.

    E. sometimes increases when new variables are added.

13. Consider the following regression model:

$$y = \beta_0 + \beta_1 x_1 + u$$

    If the sample size is $n = 684$, then the degrees of freedom is dof = ...

    A. 684

    B. 683

    C. 685

    D. 2

    E. 682

14. Suppose that we run the regression of college GPA on high school GPA and obtain $R^2 = 0.2$. What is the correct interpretation of the coefficient of determination?

    A. Approximately 80% of the variations in the high school GPA can be explained by the variations in college GPA.

    B. Approximately 80% of the variations in the college GPA can be explained by the variations in high school GPA.

    C. Approximately 20% of the variations in the high school GPA can be explained by the variations in college GPA.

    D. Approximately 20% of the variations in the college GPA can be explained by the variations in high school GPA.

    E. Cannot be found from the given information.

15. **Consider the following (edited) R output:**

```
Call:
lm(formula = final ~ attend + priGPA + ACT, data = attend)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.83420    1.42371   6.907 1.14e-11
attend       0.07934    0.03523   2.252   0.0247
priGPA       1.91529    0.37261   5.140 3.60e-07
ACT          0.40106    0.05323   7.535 1.57e-13
---
Residual standard error: 4.218 on 676 degrees of freedom
Multiple R-squared:  0.2013,Adjusted R-squared:  0.1978
F-statistic: 56.79 on 3 and 676 DF,  p-value: < 2.2e-16
```

In the model above, final is the final scores from a class, attend is the number of classes attended by the student (out of 32), priGPA is the prior GPA, and ACT is the score from the university entrance exam.

Which of the following is correct?

A.

$$\widehat{\text{final}} = \underset{(1.42)}{9.83} + \underset{(0.035)}{0.079}\,\text{attend} + \underset{(0.373)}{1.915}\,\text{priGPA} + \underset{(0.053)}{0.401}\,\text{ACT}$$
$$n = 676 \quad R^2 = 0.1978$$

B.

$$\widehat{\text{final}} = \underset{(1.42)}{0.20} + \underset{(0.035)}{0.079}\,\text{attend} + \underset{(0.373)}{1.915}\,\text{priGPA} + \underset{(0.053)}{0.401}\,\text{ACT}$$
$$n = 600 \quad R^2 = 0.1978$$

C.

$$\widehat{\text{final}} = \underset{(1.42)}{9.83} + \underset{(0.035)}{0.079}\,\text{attend} + \underset{(0.373)}{1.915}\,\text{priGPA} + \underset{(0.053)}{0.401}\,\text{ACT}$$
$$n = 680 \quad R^2 = 0.2013$$

D.

$$\widehat{\text{final}} = \underset{(1.42)}{9.83} + \underset{(0.035)}{0.079}\,\text{attend} + \underset{(0.373)}{1.915}\,\text{priGPA} + \underset{(0.053)}{0.401}\,\text{ACT}$$
$$n = 680 \quad R^2 = 0.5679$$

E.

$$\widehat{\text{final}} = \underset{(1.42)}{9.83} + \underset{(0.035)}{0.079}\,\text{attend} + \underset{(0.373)}{1.915}\,\text{priGPA} + \underset{(0.053)}{0.401}\,\text{ACT}$$
$$n = 676 \quad R^2 = 0.5679$$